

SC4016: CYBER THREAT INTELLIGENCE

GROUP ASSIGNMENT

GROUP 13

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ACADEMIC YEAR 2026/2027

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Introduction

Ransomware groups use data leak sites (DLS) to name victims who have not paid and to publish their stolen data. We studied five DLS: **Anubis, Chaos, DireWolf, Play and Settra**, and collected every victim posted between 1 January and 30 September 2026.

Cross-referencing

Chaos DLS shows no post dates, so dates were taken from ransomware.live. Country, industry and revenue were verified from company websites and business directories. Aggregators were used only for cross-checking, never as the source of victims. External reports dated before 2026 are used only to describe how the groups operate (TTPs) or for general background, never as victim data.

Dataset

The dataset contains 398 victims posted on 5 data leak sites, plus 48 posts that were still on countdown at the time of scraping (not included in the analysis).

Group	Victims	Share	First post	Last post	Posting days	Countries
Play	167	42.0%	2 Jan	21 Sep	58	11
Settra	68	17.1%	2 Jun	23 Sep	21	17
Anubis	65	16.3%	8 Jan	22 Sep	52	15
Chaos	55	13.8%	5 Jan	29 Sep	41	4
DireWolf	43	10.8%	4 Jan	8 Sep	12	19
Total	**398**	100%	2 Jan	29 Sep		38

Table 1: Dataset summary

Field	Play	Settra	Anubis	Chaos	DireWolf
Country (known)	167	67	57	53	43
Industry (known)	154	68	65	55	40
Revenue band	0	68	0	0	0
Data types (specific)	0	68	45	0	32
Claimed leak size	0	63	48	55	34

Table 2: Field coverage by group (number of victims with the field filled)

Answers

Q1. Most targeted countries by ransomware actors

- Victims were located in 38 countries; 11 victims (2.8%) have no identified country.
- The United States accounts for 262 victims (65.8%). The US and Canada together account for 291 victims (73.1%).
- The next most targeted countries are Canada (29, 7.3%), the United Kingdom (21, 5.3%), Germany (9, 2.3%) and France (6, 1.5%).

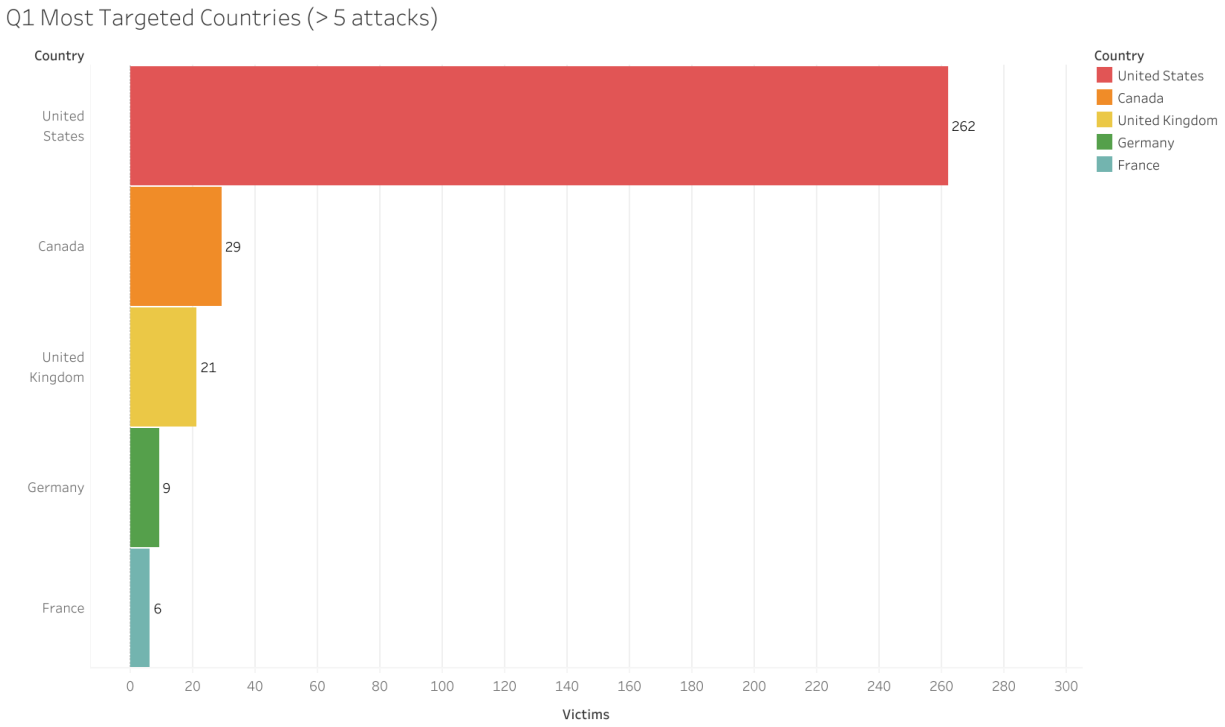


Diagram 1: Bar chart of the most targeted countries

Q2. Why are some countries more targeted than others?

- DireWolf is the only group where US victims are a minority (30.2%); its victims span the most countries (19).
- Chaos has the narrowest spread: 4 countries, with 92.7% of victims in the US or Canada.
- Asia-Pacific victims (excluding Australia/NZ): 14 in total: Malaysia 4 (DireWolf), India 3 (DireWolf 2, Settra 1), South Korea 2 (Anubis), and 1 each in China (Anubis), Taiwan (Chaos), Thailand (DireWolf), Japan (Settra) and Singapore (Settra).
- Latin America: 10 victims (Colombia 2, Brazil 2, Chile, Peru, Argentina, Guatemala, Ecuador, Mexico).
- No victims were located in Russia or any other member states of the Commonwealth of Independent States.

Based on the gathered data, we concluded that these may be the reason why some countries are more targeted than others.

1. Exposed remote-access infrastructure. Play's documented entry points are internet-facing systems: Fortinet FortiOS and Microsoft Exchange vulnerabilities, plus RDP and VPN accounts. Settra also gains entry through compromised VPN credentials. In June 2026, during our study period, Singapore's Cyber Security Agency (CSA) warned that over 70,000 FortiGate device credentials had been compromised through brute-force and credential-stuffing attacks, which could give attackers VPN access and a route into the internal network. Economies with very large numbers of small and mid-sized firms running such devices therefore offer the largest pool of easy targets.
2. Language-dependent social engineering. As noted in Q1, phone-based intrusion favours English-speaking countries. CSA and the Singapore Police Force also warned in June 2026 about technical-support scams impersonating Microsoft, showing that this style of deception remained active during the study period.
3. Deliberate avoidance of CIS countries. Our dataset contains no victims in Russia or other CIS countries. Blockchain-analysis research reported that over 90% of ransom payments since 2020 went to ransomware strains designed to avoid CIS and Russian-speaking countries, often by checking for a Cyrillic keyboard. The absence of CIS victims in all five groups is consistent with this long-standing pattern.
4. Ability to pay. Wealthier economies host more companies able to pay large ransoms. Within our data, Settra's claimed leak sizes rise with victim revenue, indicating that larger firms are also worth more to the attacker.
5. Visibility bias. A DLS only shows victims who have yet to pay, so countries where victims tend to pay quickly may be under-represented in leak-site data.
6. Rule among ransomware actors. None of the victims is in Russia or other CIS countries, consistent with the widely reported rule in Russian-speaking ransomware programs not to attack the CIS.

Q3. Most prolific or successful ransomware group

- Play posted the most victims: 167 (42.0%).
- Settra was only active from June. Per active month, Settra (17.0 victims/month over 4 months) is close to Play (18.6/month over 9 months).
- Play and Anubis both posted new victims in 32 of the 39 weeks; Chaos in 26; Settra in 11; DireWolf in 8.

Measure	Play	Settra	Anubis	Chaos	DireWolf
Victims	**167**	68	65	55	43
Months with posts	9	4 (Jun–Sep)	9	9	4 (Jan, Jun, Aug, Sep)
Victims per active month	**18.6**	17.0	7.2	6.1	10.8
Posting days	58	21	52	41	12
Avg victims per posting day (max)	2.9 (9)	3.2 (9)	1.3 (3)	1.3 (4)	**3.6 (7)**
Weeks with posts (of 39)	**32**	11	**32**	26	8

Median claimed leak size	not disclosed	144 GB	199 GB	**400 GB**	66 GB
Total claimed data	not disclosed	21.3 TB	19.1 TB	**28.2 TB**	6.7 TB
Largest single claim	not disclosed	3.2 TB	**3.6 TB**	3.2 TB	1.15 TB
Posts on countdown at snapshot	0	7	0	**41**	0

Table 3: Activity and volume measures

Play is the most prolific group by every volume measure: total victims, victims per month and consistency across the nine months. If we were to include external reports dated before 2026 to back this finding, the scale reported by US and Australian authorities, who counted about 900 organisations affected by Play by May 2025. Play operates as a closed group rather than an affiliate program and recompiles its malware for every attack so that each sample has a different file hash, which defeats signature-based antivirus. A tightly controlled, repeatable operation fits the steady posting pattern seen in our data.

Settra is the fastest-rising group. Although it was first posted in June 2026, its monthly rate (17.0) almost matches Play's. Settra is described as a new operation that emerged in June 2026 with no confirmed links to earlier groups. Its total is therefore not comparable to groups active all year.

Chaos claims the largest thefts. Its median claimed leak (400 GB) is the highest of the five groups and it had 41 posts on countdown at once when scraped, more than all other groups combined. Cisco Talos observed Chaos demanding ransoms of up to US$300,000 and threatening DDoS attacks and notification of clients and competitors to increase pressure.

Q4. Unique ransomware actors' TTPs

DLS posts do not describe attack methods, so this section is drawn entirely from external threat reporting.

Group	Initial access	Execution and evasion	Encryption / impact	Extortion
Play (CISA et al., 2025; The Hacker News, 2025a)	Exploits known flaws in FortiOS (CVE-2018-13379, CVE-2020-12812), Exchange (CVE-2022-41040/41082) and SimpleHelp (CVE-2024-57727); valid RDP/VPN accounts; also used Windows	AdFind, Grixba, Cobalt Strike, SystemBC, PsExec, Mimikatz; disables antivirus; new binary compiled per victim	Intermittent encryption; ESXi variant for virtual machines	Closed group; email contact; phone calls to pressure victims

	zero-day CVE-2025-29824			
Anubis (Policarpio et al., 2025)	Spear-phishing emails	Checks for admin rights; deletes shadow copies	Optional /WIPEMODE permanently destroys files	RaaS with affiliate streams for encryption, data-only extortion and access sales
Chaos (Bennett et al., 2025)	Spam flooding, then phone call posing as IT support; victim opens Microsoft Quick Assist	Legitimate remote tools (AnyDesk, ScreenConnect, Splashtop); GoodSync for data theft	Fast, selective, multi-threaded encryption	Demands up to US$300K; threats of DDoS and of notifying clients/competitors
Dire Wolf (CSA, 2025; Halcyon, 2025)	Not confirmed; suspected phishing and exposed RDP/VPN	Stops about 75 security and backup services; kills event logging; deletes shadow copies and backups	Curve25519 + ChaCha20; files over 1 MB only partly encrypted	Double extortion; demands up to US$500K; 30-day negotiation deadline
Settra (MOXFIVE, 2026; Proven Data, 2026)	Compromised VPN credentials, often from infostealer infections	NetExec, Mimikatz, PAExec, Mesh Agent; disables EDR via vulnerable driver; clears event logs	Encryption not yet documented; behaves as a "data broker"	Long-form "exposé" leak posts on Tor site

Table 4: Tactics, techniques, and procedures of each threat actor

Similar features are observed among the 5 threat actors:

1. Most access comes from known but unpatched vulnerabilities, stolen or weak credentials, and people rather than technical exploits.
2. All five rely on legitimate admin and remote-access tools, and several disable security software or clear logs, which makes detection harder.
3. Speed, intermittent encryption and partial encryption of large files lock systems is done in a fraction of the time full encryption would take.
4. Layered pressure. Data theft is combined with encryption, file destruction, backup deletion and direct harassment. Our data shows the follow-through: Play published 66.5% of its victims exactly four days after listing and 86.8% in full.

Q5. Threat actors to “re-brand” or “repurpose” older data leaks to inflate reputation

- No victim was posted by more than one group. Checked by matching domains and normalised victim names across all 398 records.
- No group posted the same victim twice under the same name or domain.
- One victim was explicitly re-posted: Anubis listed a New Zealand law firm as "Part 2" on 9 February 2026 (1 of 398 records, 0.3%).

Based on our gathered data, reposting old leaks is rare. The only repost is a second release of data from the same attack. Each listing appears to represent a distinct intrusion. However, rebranding of the group themselves is common. Chaos is assessed to be former BlackSuit members, and BlackSuit was itself a rebrand of Royal. Anubis began as a development-stage variant called Sphinx. Settra may be a new operation or a re-branded affiliate cluster. Dire Wolf was assessed as a new actor with its own infrastructure. Groups change names after law-enforcement pressure or a damaged reputation, but they keep their tools and people.

Q6. Victim company size or revenue correlation to risk of attacks

Revenue is only available for Settra's 68 victims (taken from Settra's own posts). None of the other four DLS publish revenue. Employee counts are available for 3 victims only.

Revenue band	Victims	% of 68	Median claimed leak size
< $10M	25	36.8%	68 GB
$10M – $50M	23	33.8%	150 GB
$50M – $250M	14	20.6%	255 GB
$250M – $1B	4	5.9%	700 GB
> $1B	2	2.9%	2,285 GB

Table 5: Settra victims by revenue band

- 70.6% of Settra victims reported revenue under US$50M; 8.8% above US$250M.
- Median claimed leak size rises with each revenue band, from 68 GB (< $10M) to 2,285 GB (> $1B).
- Financial Services & Insurance accounts for 11 of the 25 victims under $10M.

Smaller companies make up most of Settra's victims, suggesting that smaller firms face a higher risk of being attacked. Small and mid-sized firms are far more numerous and typically have fewer security staff. Access methods based on stolen credentials and unpatched devices (Q4) succeed most often against exactly these organisations.

Larger companies are higher-impact victims: claimed leak size rises steadily with revenue band, from a median of 68 GB to 2,285 GB. Larger firms hold more data and can pay more, so each successful attack is worth more to the attacker.

Based on our data we are unable to conclude if the company size has any correlation with ransomware attack and should be treated as indicative.

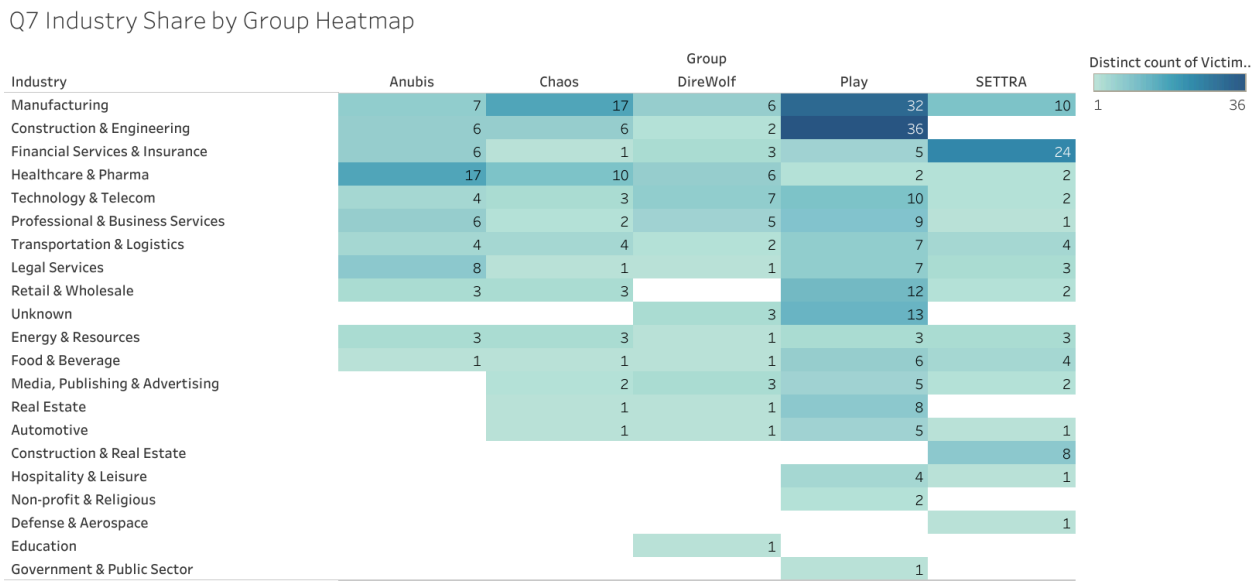
Q7. Which Industries Are More Prone To Ransomware Threats? Why?

Based on data: Manufacturing 72 (18.1%), Construction & Engineering 50 (12.6%), Financial Services 39 (9.8%), Healthcare 37 (9.3%). The largest industry differs for each threat group.

1. 16 victims (4.0%) have no identified industry (Play 13, DireWolf 3).
2. Manufacturing. Downtime halts production and revenue, creating strong pressure to pay. Sophos (2025) found that 51% of manufacturing victims paid the ransom and that exploited vulnerabilities were the most common root cause, matching the vulnerability-driven access used by Play, whose victims are 19% manufacturing.

3. Construction & Engineering is almost entirely a Play target (36 of 50 victims). These firms often run internet-facing remote-access systems for distributed project sites with limited IT staff, which matches Play's reliance on exposed VPN, RDP and edge devices.
4. Healthcare is targeted mainly by Anubis (26% of its victims), Chaos (18%) and DireWolf (14%). US health officials describe healthcare as vulnerable because of complex, interconnected hospital infrastructure, historically underfunded cybersecurity and many legacy systems. Patient records also give attackers strong leverage, which suits Anubis's data-extortion affiliate model.
5. Financial Services is 35% of Settra victims and 9% or less for every other group. Settra operates as a "data broker" that publishes investigative reports. Financial firms hold the sensitive client and financial records that make such exposure damaging.

Overall, industry risk depends on each group-level specialisation.



Industry	Anubis	Chaos	DireWolf	Play	SETTRA
Manufacturing	7	17	6	32	10
Construction & Engineering	6	6	2	36	
Financial Services & Insurance	6	1	3	5	24
Healthcare & Pharma	17	10	6	2	2
Technology & Telecom	4	3	7	10	2
Professional & Business Services	6	2	5	9	1
Transportation & Logistics	4	4	2	7	4
Legal Services	8	1	1	7	3
Retail & Wholesale	3	3		12	2
Unknown			3	13	
Energy & Resources	3	3	1	3	3
Food & Beverage	1	1	1	6	4
Media, Publishing & Advertising		2	3	5	2
Real Estate		1	1	8	
Automotive		1	1	5	1
Construction & Real Estate					8
Hospitality & Leisure				4	1
Non-profit & Religious				2	
Defense & Aerospace					1
Education			1		
Government & Public Sector				1	

Diagram 2: Heatmap to depict the most commonly targeted industries

Q8. What kind of data do actors usually target? What are the kinds of data targeted in each industry?

- Specific data types are available for 145 victims: Settra 68, Anubis 45, DireWolf 32. Play (167) and Chaos (55) posts did not state data types; a further 31 Anubis/DireWolf posts said data was stolen without naming the type.
- Settra posts list more data types per victim (average 6.1) than DireWolf (3.2) or Anubis (1.7), so Settra makes up the majority of tagged data types.
- Settra's labels were mapped to the shared categories for this table (e.g. "Financial records" to Financial, "PII (SSN/ID/passport)" to PII, "Contracts & legal" plus Legal and Contracts to Legal & Contracts). Emails and Insurance records appear only in Settra's data.

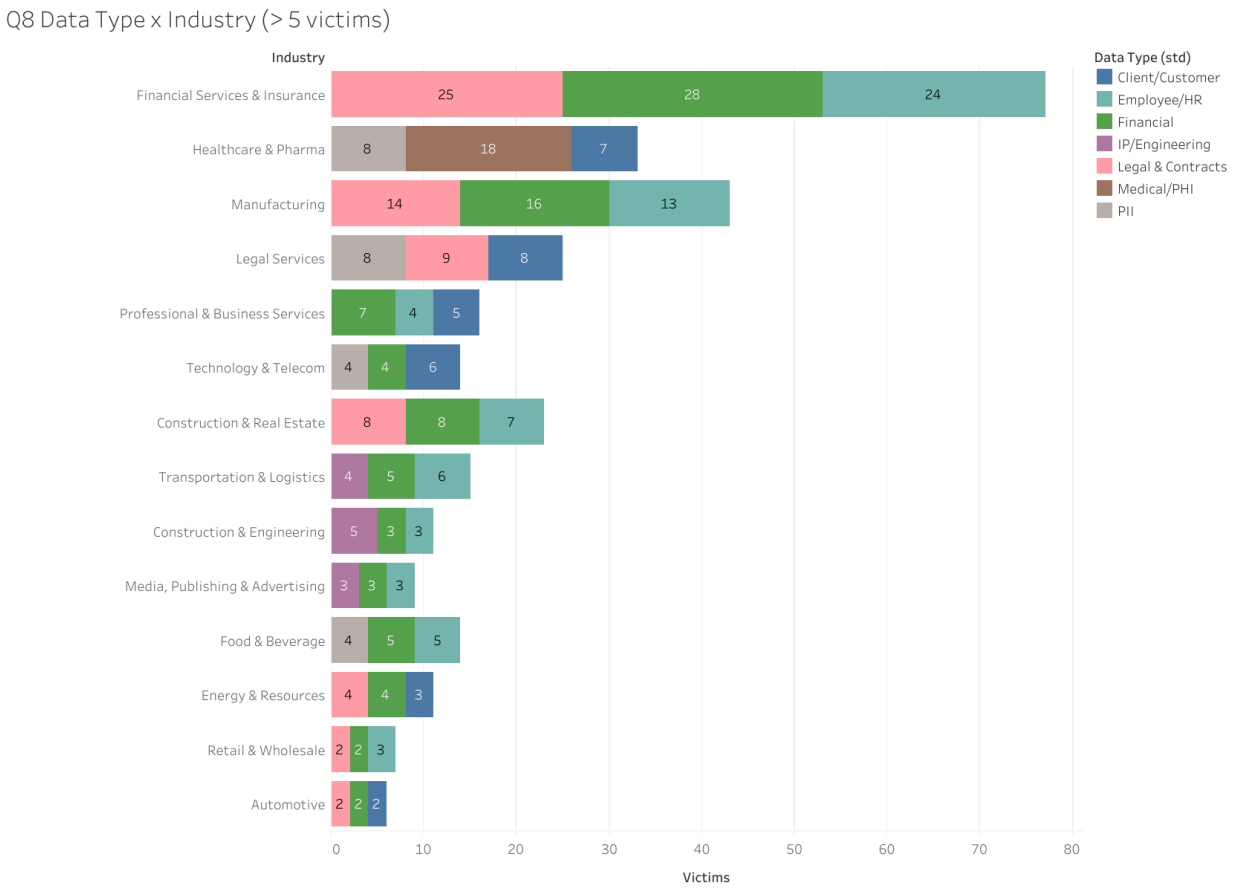


Diagram 3: Data types targeted from victims in each industry.

Actors most often target financial records, employee/HR files and contracts, data that every organisation holds and that is sensitive regardless of industry. Personal identifiers (security numbers, passports, dates of birth) are also common because they enable fraud and trigger legal notification duties.

Beyond these, attackers take the data that is most damaging for each industry: patient records in healthcare, designs and project files in construction and engineering, client and case files in legal services, and customer data in technology. Threat actors are copying only files modified in the past year and excluding large or unusual file types, focusing on usable data.

Q9. Share 3 interesting insights you observed.

The first interesting insight that we observed was that DireWolf changed from encrypting companies to stealing cloud databases.

DireWolf's 16 posts from January to June look like classic ransomware victims such as manufacturers and engineering firms. However from August, 22 of its 27 posts were database dumps with row counts, for example about 1.28 billion rows from cartrack and about 283.6 million records from Aztec Software. Almost every victim is an online platform like fintech wallets and a university.

The second insight was that the data published on leak sites cannot be taken at face value.

Play's view counter behaves like a clock, with its post's age predicting its views almost perfectly. A victim post shows about 43000 views and a September post about 3000, however well known the victim is. The views come from crawlers or are inflated and not from public interest.

Lastly, the third insight is that every employer is a health-data holder.

Settra claims medical or health data at 38 of its 66 victims that are not healthcare firms. The post shows where it comes from, HR files such as sick notes, workers' compensation claims and insurance forms. A ski-resort operator or lumber yard can therefore face health-data breach obligations. Security teams outside healthcare rarely treat HR files as sensitive yet attackers clearly do.

Q10. Lessons learnt, struggles in executing the project and how did you overcome them?

Lessons Learnt

The project has taught us that researching ransomware groups and collecting data from dark web leak sites is complex. Each ransomware group is unique in their website structure. Thus, there was no single scraper that could work for every site. Some sites used HTML tables, JSON etc. while some had missing or inconsistent information. We overcame this by inspecting the page structure manually, adjusting selectors, creating site-specific methods and testing our unique scripts on smaller samples before obtaining the full dataset.

From our analysis, ransomware groups operate uniquely. Different groups showed different preferences for countries, industries and victim sizes. For example, Play had the highest number of victims in our dataset while DireWolf had a wider geographic spread. Ransomware at this state is increasingly focused on data theft and extortion rather than only encrypting systems. Commonly exposed information involved in financial records, personal data and business documents.

Overall, this project gave us practical experience in cyber threat intelligence, dark web research, web scraping, data cleaning and validation.

Struggles

A major challenge was the unstable nature of Tor and Onion services where pages could time out, disappear or return incomplete responses. We then limited our collection to victim metadata and avoided downloading or interacting with leaked data. Information published by ransomware groups should be treated as claims and validated using legitimate public sources.

Solutions

To reduce data loss, we added retries, error handling and status fields so that failed/ incorrectly parsed pages could be identified and checked again. For some scripts, data was saved progressively to ensure any previously collected results are not lost when a connection failure occurs. Digital Safety is an important lesson. As we had limited prior experience working with Tor and leak sites, we utilised an isolated Kali VM, snapshots, disabled file sharing and careful browsing practices.

A key improvement we would make in the future would be to design a common scraping framework from the start with shared error handling, Tor verification, deduplication and a standard output format for all ransomware groups.

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